

CSE 5825: Probabilistic Graph Analysis

Homework 1

Problem 1 (10 points)

Prove

$$p(x, y | z) = p(x | z) p(y | x, z)$$

and also

$$p(x | y, z) = \frac{p(y | x, z) p(x | z)}{p(y | z)}.$$

First identity. We can apply the ordinary product rule to get $p(u, v) = p(u | v) p(v)$, inside the world where z is already given – i.e. treat $p(\cdot | z)$ as an ordinary distribution and apply the product rule to x and y within it, conditioning on x first:

$$p(x, y | z) = p(x | z) p(y | x, z).$$

Second identity. Apply the product rule again to the same quantity $p(x, y | z)$, but conditioning on y first instead:

$$p(x, y | z) = p(y | z) p(x | y, z).$$

The left-hand sides of the two expansions are identical, so the right-hand sides must be equal to each other:

$$p(x | z) p(y | x, z) = p(y | z) p(x | y, z).$$

Dividing both sides by $p(y | z)$ (assuming $p(y | z) \neq 0$) and solving for $p(x | y, z)$ gives

$$p(x | y, z) = \frac{p(y | x, z) p(x | z)}{p(y | z)},$$

which is exactly the second identity. ■

Problem 2 (15 points)

A college has three departments, A, B, and C, with female percentages 35%, 20%, and 55% respectively. A department is chosen at random, then a student is chosen at random from that department.

(a) $P(\text{female})$

$$P(\text{female}) = P(\text{female} | \text{Dept} = A)P(\text{Dept} = A) + P(\text{female} | \text{Dept} = B)P(\text{Dept} = B) + P(\text{female} | \text{Dept} = C)P(\text{Dept} = C) = 0.35 * \frac{1}{3} + 0.20 * \frac{1}{3} + 0.55 * \frac{1}{3} = \frac{0.35+0.20+0.55}{3} = \frac{1.10}{3} \approx 0.3667.$$

(b) $P(\text{Dept} = B \mid \text{female})$

$$P(\text{Dept} = B \mid \text{female}) = \frac{P(\text{female} \mid \text{Dept}=B)P(\text{Dept}=B)}{P(\text{female})} = \frac{0.20 \times \frac{1}{3}}{0.3667} \approx 0.1818.$$

(c) $P(\text{Dept} = A \mid \text{male})$

$$P(\text{Dept} = A \mid \text{male}) = \frac{(1-0.35) \times \frac{1}{3}}{1-0.3667} = \frac{0.65 \times 0.3333}{0.6333} = \frac{0.2167}{0.6333} \approx 0.3421$$

Problem 3 (20 points)

Consider $P(A, B, C, D)$ where A, B are binary and C, D have 3 values each.

(a) **Parameters for the full joint**

The product of the number of values for each variable is $2 * 2 * 3 * 3 = 36$. Since the probabilities must sum to 1, we have $36 - 1 = 35$ free parameters.

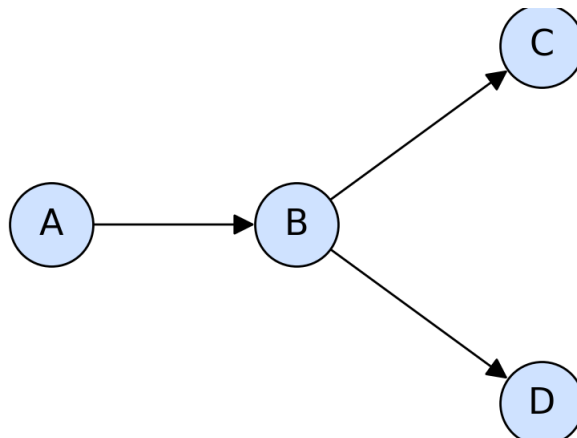
(b) **Parameters assuming $P(A, B, C, D) = P(A)P(B \mid A)P(C \mid B)P(D \mid B)$**

We know that $P(X \mid \text{Parents})$ costs $r * (k - 1)$ parameters, where $k = \# \text{values of } X$ and $r = \text{product of } \# \text{values of each parent}$, so we can compute the cost for each factor as:

- Factor $P(A)$: $k = 2$ (A is binary), parents = none so $r = 1 \rightarrow \text{cost} = 1 * (2 - 1) = 1$.
- Factor $P(B \mid A)$: $k = 2$ (B is binary), parents = $\{A\}$ so $r = 2$ (A has 2 values) $\rightarrow \text{cost} = 2 * (2 - 1) = 2$.
- Factor $P(C \mid B)$: $k = 3$ (C has 3 values), parents = $\{B\}$ so $r = 2$ (B has 2 values) $\rightarrow \text{cost} = 2 * (3 - 1) = 4$.
- Factor $P(D \mid B)$: $k = 3$ (D has 3 values), parents = $\{B\}$ so $r = 2$ (B has 2 values) $\rightarrow \text{cost} = 2 * (3 - 1) = 4$.

Total parameters = $1 + 2 + 4 + 4 = 11$.

(c) **Network diagram and independence questions**



Is $A \perp B$? Is $A \perp B \mid C$? A is not independent of B , since there is a direct edge $A \rightarrow B$. A is not independent of B given C , since conditioning on C does not block the direct edge $A \rightarrow B$.

Problem 4 (30 points)

Chest Clinic belief network. Variables: x = positive X-ray, d = dyspnea, e = either tuberculosis or lung cancer, t = tuberculosis, l = lung cancer, b = bronchitis, a = visited Asia, s = smoker.

(a) Conditional independence statements (true/false)

1. tuberculosis $\perp$ smoking $\mid$ shortness of breath No, there is an open path $t-e-l-s$.
2. lung cancer $\perp$ bronchitis $\mid$ smoking No, there is an open path $l-e-d-b$.
3. visit to Asia $\perp$ smoking $\mid$ lung cancer No, there is an open path $a-t-e-d-b-s$.
4. visit to Asia $\perp$ smoking $\mid$ lung cancer, shortness of breath Yes, all paths between a and s are blocked given l and d .

(b) $p(d)$, $p(d \mid s = \text{tr})$, $p(d \mid s = \text{fa})$

For this question I used the following factorization of the joint distribution, which is consistent with the network structure:

$$p(a, t, s, l, b, e, x, d) = p(a) p(t \mid a) p(s) p(l \mid s) p(b \mid s) p(e \mid t, l) p(x \mid e) p(d \mid e, b).$$

Algorithm 1 Marginal and conditional queries on the Chest Clinic network

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1: procedure JOINTPROB( $a, t, s, l, b, e, x, d$ )
2:   return  $p(a) \cdot p(t \mid a) \cdot p(s) \cdot p(l \mid s) \cdot p(b \mid s) \cdot p(e \mid t, l) \cdot p(x \mid e) \cdot p(d \mid e, b)$ 
3: end procedure
4: procedure MARGINALD
5:    $p_d[\text{tr}] \leftarrow 0$ ,    $p_d[\text{fa}] \leftarrow 0$ 
6:   for each assignment of  $a, t, s, l, b, e, x, d$  do
7:      $p_d[d] \leftarrow p_d[d] + \text{JOINTPROB}(a, t, s, l, b, e, x, d)$ 
8:   end for
9:   return  $p_d$ 
10: end procedure
11: procedure CONDITIONALDGIVENS( $s^*$ )
12:    $u[\text{tr}] \leftarrow 0$ ,    $u[\text{fa}] \leftarrow 0$ 
13:   for each assignment of  $a, t, s, l, b, e, x, d$  with  $s = s^*$  do
14:      $u[d] \leftarrow u[d] + \text{JOINTPROB}(a, t, s, l, b, e, x, d)$ 
15:   end for
16:   return  $u$  normalized so  $u[\text{tr}] + u[\text{fa}] = 1$ 
17: end procedure

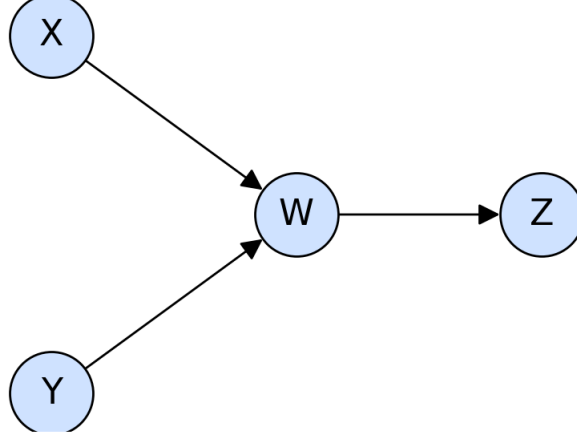
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$$\begin{aligned}
p(d) &= 0.4359706 \\
p(d \mid s = \text{tr}) &= 0.552808 \\
p(d \mid s = \text{fa}) &= 0.31913319999999995
\end{aligned}$$

Problem 5 (15 points)

Consider $p(x, y, w, z) = p(z | w) p(w | x, y) p(x) p(y)$.

(a) Network diagram



(b) $p(x | z)$

$$p(x | z) = \frac{p(x, z)}{p(z)} = \frac{p(x) \sum_y \sum_w p(z | w) p(w | x, y) p(y)}{\sum_w p(z | w) \sum_{x'} \sum_y p(w | x', y) p(x') p(y)}$$

(c) $p(y | z)$

$$p(y | z) = \frac{p(y, z)}{p(z)} = \frac{p(y) \sum_x \sum_w p(z | w) p(w | x, y) p(x)}{\sum_w p(z | w) \sum_{y'} \sum_x p(w | x, y') p(x) p(y')}$$

(d) Condition for $x \perp y | z$

$$x \perp y | z = p(x | y, z) = p(x | z) = \frac{p(x, z)}{p(z)} = \frac{p(x) \sum_y \sum_w p(z | w) p(w | x, y) p(y)}{\sum_w p(z | w) \sum_{x'} \sum_y p(w | x', y) p(x') p(y)}$$

Problem 6 (15 points)

Graph: $A \rightarrow C, A \rightarrow E, B \rightarrow C, B \rightarrow D, C \rightarrow E$.

(a) Independent pairs, no observations

The independent pairs with no observations are:

- (A, B) : The only paths $(A \rightarrow C \leftarrow B$ and through $E)$ pass through the collider C , which is unobserved, so A and B are independent.
- (A, D) : Every path from A to D also passes through the same collider C (then $B \rightarrow D$), which is unobserved, so A and D are independent.

(b) Independent pairs given E

There are no independent pairs given E , since observing E (a descendant of the collider at C) unblocks that collider, making both (A, B) and (A, D) dependent.

(c) Decomposition of $P(A, B, C, E)$

$$P(A, B, C, E) = P(A)P(B)P(C \mid A, B)P(E \mid A, C)$$

(d) Parameters for CPD of E (A, B, D binary; C, E three values)

E 's parents are A (binary, 2 values) and C (3 values), so $r = 2 \times 3 = 6$ and $k = 3$ (E 's own values). The parameters for CPD of E is $r * (k - 1) = 6 * (3 - 1) = 12$.